Pre-Calculus Concepts

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Limits: an introductory numerical approach

Estimate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.

Factor: $\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$. As $x \to 2$, this approaches $2 + 2 = 4$.

2 [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = x^2 + 1$ over the interval $[1, 4]$.

Average rate of change $= \frac{f(4) - f(1)}{4 - 1} = \frac{17 - 2}{3} = \frac{15}{3} = 5$.

3 [Foundational] • Target: Conic sections: ellipses

Find the equation of an ellipse centered at the origin with $a = 5$ (horizontal semi-axis) and $b = 3$ (vertical semi-axis).

$\frac{x^2}{25} + \frac{y^2}{9} = 1$.

4 [Foundational] • Target: Conic sections: ellipses

For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.

Since $16 > 9$, the major axis is horizontal, with $a = \sqrt{16} = 4$. Vertices: $(4, 0)$ and $(-4, 0)$.

5 [Foundational] • Target: Conic sections: hyperbolas

Find the equation of a hyperbola centered at the origin with vertices at $(\pm 4, 0)$ and $b = 3$.

Since the vertices are horizontal, the equation has the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, with $a = 4$: $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

6 [Foundational] • Target: Vectors in the plane

Given $\mathbf{u} = (3, -4)$, find its magnitude and a unit vector in the same direction.

$|\mathbf{u}| = \sqrt{9 + 16} = 5$. Unit vector: $\frac{1}{5}(3, -4) = (0.6, -0.8)$.

7 [Foundational] • Target: A multi-part conic and function analysis problem

This question has two parts. A circle has equation $x^2 + y^2 - 4x + 6y - 12 = 0$. (a) Complete the square to write the equation in standard form, and state the center and radius. (b) Determine whether the point $(6, -1)$ lies inside, on, or outside the circle.

(a) Group and complete the square: $(x^2 - 4x + 4) + (y^2 + 6y + 9) = 12 + 4 + 9$, giving $(x - 2)^2 + (y + 3)^2 = 25$. Center: $(2, -3)$. Radius: 5. (b) Distance from center to $(6, -1)$: $\sqrt{(6 - 2)^2 + (-1 - (-3))^2} = \sqrt{16 + 4} = \sqrt{20} \approx 4.47$. Since $4.47 < 5$, the point lies INSIDE the circle.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Piecewise and step functions in applied contexts

A shipping company charges 5 dollars per pound for the first 10 pounds, then 3 dollars per pound for each pound beyond 10. Write a piecewise function $C(w)$ for the cost, and find the cost of a 15-pound package.

$C(w) = 5w$ for $0 < w \leq 10$, and $C(w) = 50 + 3(w - 10)$ for $w > 10$. For 15 pounds: $C(15) = 50 + 3(5) = 50 + 15 = 65$ dollars.

9 [Intermediate] • Target: A multi-part conic section identification and analysis problem

This question has three parts. An equation is given: $4x^2 + 9y^2 - 16x + 18y - 11 = 0$. (a) Complete the square in both x and y to write the equation in standard conic form. (b) Identify the type of conic section. (c) State the center and the lengths of the semi-axes.

(a) Group terms: $4(x^2 - 4x) + 9(y^2 + 2y) = 11$. Complete the square: $4(x^2 - 4x + 4) + 9(y^2 + 2y + 1) = 11 + 16 + 9$, giving $4(x - 2)^2 + 9(y + 1)^2 = 36$. Divide by 36: $\frac{(x-2)^2}{9} + \frac{(y+1)^2}{4} = 1$. (b) Since both terms are positive and added (not subtracted) with different denominators, this is an ellipse. (c) Center: $(2, -1)$. Semi-axes: $a = \sqrt{9} = 3$ (horizontal), $b = \sqrt{4} = 2$ (vertical).

10 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Estimate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.

Factor: $\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$. As $x \to 2$, this approaches $2 + 2 = 4$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.

Since $16 > 9$, the major axis is horizontal, with $a = \sqrt{16} = 4$. Vertices: $(4, 0)$ and $(-4, 0)$.

12 [Intermediate] • Target: Parametric equations

Eliminate the parameter from $x = t + 1$ and $y = t^2$, expressing y in terms of x.

From the first equation, $t = x - 1$. Substitute: $y = (x - 1)^2$.

13 [Intermediate] • Target: Sequences and series: sigma notation

Evaluate $\sum_{k=1}^{5} (2k + 1)$.

$\sum_{k=1}^{5} (2k + 1) = 3 + 5 + 7 + 9 + 11 = 35$.

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14 [Intermediate] • Target: Sequences and series: sigma notation

Write $4 + 8 + 12 + 16 + 20$ using sigma notation, and evaluate it.

This is $\sum_{k=1}^{5} 4k$. Evaluating: $4(1 + 2 + 3 + 4 + 5) = 4(15) = 60$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Binomial theorem • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Use the binomial theorem to expand $(x + 2)^4$.

$(x + 2)^4 = x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16 = x^4 + 8x^3 + 24x^2 + 32x + 16$.

16 [Challenging] • Target: Binomial theorem • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the coefficient of x^3 in the expansion of $(x + 3)^5$, using $\binom{5}{k}$.

The term with x^3 comes from $\binom{5}{3}x^3(3)^2 = 10x^3(9) = 90x^3$. The coefficient is 90.

17 [Challenging] • Target: Matrix operations • Cross-Lesson: 8.01 Statistics, Counting & Probability

Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find $A - B$.

Subtract corresponding entries: first row $(2 - 0, -1 - 5) = (2, -6)$; second row $(3 - (-2), 4 - 1) = (5, 3)$.

18 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Advanced Number Theory

Find the sum of the infinite geometric series $8 + 4 + 2 + 1 + \cdots$, using $S = \frac{a_1}{1 - r}$.

First term $a_1 = 8$, ratio $r = \frac{1}{2}$. Sum $= \frac{8}{1 - \frac{1}{2}} = \frac{8}{\frac{1}{2}} = 16$.

19 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 2.01 Advanced Algebra & Systems

Write the repeating decimal $0.\overline{6}$ as a fraction, by treating it as an infinite geometric series $\frac{6}{10} + \frac{6}{100} + \frac{6}{1000} + \cdots$.

First term $a_1 = \frac{6}{10}$, ratio $r = \frac{1}{10}$. Sum $= \frac{6/10}{1 - 1/10} = \frac{6/10}{9/10} = \frac{6}{9} = \frac{2}{3}$.

20 [Challenging] • Target: Finding terms of a sequence defined recursively • Cross-Lesson: 3.01 Advanced Functions

A sequence is defined by $a_1 = 3$ and $a_n = 2a_{n-1} + 1$ for $n \geq 2$. Find a_4.

$a_2 = 2(3) + 1 = 7$. $a_3 = 2(7) + 1 = 15$. $a_4 = 2(15) + 1 = 31$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Limits: an introductory numerical approach

Find the average rate of change of $f(x) = 2x^2 - 3$ over the interval $[-3, 2]$.

a. -10

b. 0

c. -2 ✓

d. -7

$f(2) = 5$ and $f(-3) = 15$. Average rate of change $= \frac{f(2) - f(-3)}{2 + 3} = \frac{5 - 15}{5} = -2$.

22 [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = 2x^2 - 7x + 9$ over the interval $[1, 3]$.

a. 3

b. -1

c. 2

d. 1 ✓

$f(3) = 6$ and $f(1) = 4$. Average rate of change $= \frac{f(3) - f(1)}{3 - 1} = \frac{6 - 4}{2} = 1$.

23 [Foundational] • Target: Conic sections: ellipses

Find the average rate of change of $f(x) = 3x^2 + 8x - 9$ over the interval $[5, 6]$.

a. 41 ✓

b. 43

c. 40

d. 44

$f(6) = 147$ and $f(5) = 106$. Average rate of change $= \frac{f(6) - f(5)}{6 - 5} = \frac{147 - 106}{1} = 41$.

24 [Foundational] • Target: Conic sections: hyperbolas

Find the average rate of change of $f(x) = x^2 + 7x$ over the interval $[-6, 3]$.

a. 4 ✓

b. 5

c. 36

d. -5

$f(3) = 30$ and $f(-6) = -6$. Average rate of change $= \frac{f(3) - f(-6)}{3 + 6} = \frac{30 + 6}{9} = 4$.

25 [Foundational] • Target: Parametric equations

Find the average rate of change of $f(x) = x^2 - 9x + 7$ over the interval $[-3, 4]$.

a. -8 ✓

b. -56

c. -15

d. -7

$f(4) = -13$ and $f(-3) = 43$. Average rate of change $= \frac{f(4) - f(-3)}{4 + 3} = \frac{-13 - 43}{7} = -8$.

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26 [Foundational] • Target: Sequences and series: sigma notation

Find the average rate of change of $f(x) = -3x^2 + 2x + 9$ over the interval $[0, 1]$.

a. -4

b. -1 ✓

c. -2

d. 1

$f(1) = 8$ and $f(0) = 9$. Average rate of change $= \frac{f(1) - f(0)}{1} = \frac{8 - 9}{1} = -1$.

27 [Foundational] • Target: Binomial theorem

Find the average rate of change of $f(x) = x^2 - 7$ over the interval $[-2, 1]$.

a. 0

b. -3

c. -4

d. -1 ✓

$f(1) = -6$ and $f(-2) = -3$. Average rate of change $= \frac{f(1) - f(-2)}{1 + 2} = \frac{-6 + 3}{3} = -1$.

28 [Foundational] • Target: Vectors in the plane

Evaluate $\sum_{k=1}^{10} (k - 1)$.

a. 46

b. 44

c. 45 ✓

d. 55

$\sum_{k=1}^{10} (k - 1) = \sum_{k=1}^{10} k - 10 = 1(55) - 10 = 45$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A multi-part conic and function analysis problem

Evaluate $\sum_{k=1}^{10} (2k - 2)$.

a. 90 ✓

b. 88

c. 100

d. 110

$\sum_{k=1}^{10} (2k - 2) = 2\sum_{k=1}^{10} k - 20 = 2(55) - 20 = 90$.

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30 [Intermediate] · Target: Matrix operations

Evaluate $\sum_{k=1}^{10}(k+3)$.

a. 85 ✓

b. 84

c. 95

d. 55

$$\sum_{k=1}^{10}(k+3)=\sum_{k=1}^{10}k+30=1(55)+30=85.$$

31 [Intermediate] · Target: Infinite geometric series

Evaluate $\sum_{k=1}^{8}(k-5)$.

a. -5

b. 4

c. -4 ✓

d. 36

$$\sum_{k=1}^{8}(k-5)=\sum_{k=1}^{8}k-40=1(36)-40=-4.$$

32 [Intermediate] · Target: Piecewise and step functions in applied contexts

Evaluate $\sum_{k=1}^{11}(2k+1)$.

a. 132

b. 154

c. 143 ✓

d. 141

$$\sum_{k=1}^{11}(2k+1)=2\sum_{k=1}^{11}k+11=2(66)+11=143.$$

33 [Intermediate] · Target: Finding terms of a sequence defined recursively

Evaluate $\sum_{k=1}^{11}(-3k+2)$.

a. -176 ✓

b. -173

c. -165

d. -198

$$\sum_{k=1}^{11}(-3k+2)=-3\sum_{k=1}^{11}k+22=-3(66)+22=-176.$$

34 [Intermediate] · Target: A multi-part conic section identification and analysis problem

Evaluate $\sum_{k=1}^{12}(3k+5)$.

a. 306

b. 291

c. 294 ✓

d. 234

$$\sum_{k=1}^{12}(3k+5)=3\sum_{k=1}^{12}k+60=3(78)+60=294.$$

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the ellipse $\frac{x^2}{1681} + \frac{y^2}{81} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 9, 0)$

b. $(0, \pm 40)$

c. $(\pm 40, 0)$ ✓

d. $(\pm 41, 0)$

Since $a^2 = 1681$ and $b^2 = 81$: $c^2 = a^2 - b^2 = 1681 - 81 = 1600$, so $c = \sqrt{1600} = 40$. Foci: $(\pm 40, 0)$.

36 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 4, 0)$ ✓

b. $(\pm 3, 0)$

c. $(0, \pm 4)$

d. $(\pm 5, 0)$

Since $a^2 = 25$ and $b^2 = 9$: $c^2 = a^2 - b^2 = 25 - 9 = 16$, so $c = \sqrt{16} = 4$. Foci: $(\pm 4, 0)$.

37 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the ellipse $\frac{x^2}{1369} + \frac{y^2}{144} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 12, 0)$

b. $(\pm 37, 0)$

c. $(0, \pm 35)$

d. $(\pm 35, 0)$ ✓

Since $a^2 = 1369$ and $b^2 = 144$: $c^2 = a^2 - b^2 = 1369 - 144 = 1225$, so $c = \sqrt{1225} = 35$. Foci: $(\pm 35, 0)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Advanced Number Theory

A hyperbola has vertices at $(\pm 3, 0)$ and $b = 4$. Find the coordinates of the foci.

a. $(\pm 5, 0)$ ✓

b. $(\pm 7, 0)$

c. $(\pm 3, 0)$

d. $(\pm 4, 0)$

$c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Foci: $(\pm 5, 0)$.

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Advanced Algebra & Systems

A hyperbola has vertices at $(\pm 12, 0)$ and $b = 35$. Find the coordinates of the foci.

a. $(\pm 12, 0)$

b. $(\pm 35, 0)$

c. $(\pm 37, 0)$ ✓

d. $(\pm 47, 0)$

$c = \sqrt{a^2 + b^2} = \sqrt{144 + 1225} = \sqrt{1369} = 37$. Foci: $(\pm 37, 0)$.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Advanced Functions

Find the sum of the infinite geometric series with first term $a_1 = 11$ and common ratio $r = -\frac{1}{4}$, using $S = \frac{a_1}{1 - r}$.

a. $\frac{55}{4}$

b. $\frac{44}{5}$ ✓

c. -44

d. $\frac{44}{3}$

$S = \frac{a_1}{1 - r} = \frac{11}{5/4} = \frac{44}{5}$.

41 [Challenging] • Target: Limits: an introductory numerical approach • Cross-Lesson: 4.01 Advanced Trigonometry

Find the sum of the infinite geometric series with first term $a_1 = 12$ and common ratio $r = \frac{1}{4}$, using $S = \frac{a_1}{1 - r}$.

a. 9

b. 48

c. $\frac{48}{5}$

d. 16 ✓

$S = \frac{a_1}{1 - r} = \frac{12}{3/4} = 16$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

For the ellipse $\frac{x^2}{841} + \frac{y^2}{400} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 29, 0)$

b. $(0, \pm 21)$

c. $(\pm 21, 0)$ ✓

d. $(\pm 20, 0)$

Since $a^2 = 841$ and $b^2 = 400$: $c^2 = a^2 - b^2 = 841 - 400 = 441$, so $c = \sqrt{441} = 21$. Foci: $(\pm 21, 0)$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the sum of the infinite geometric series with first term $a_1 = 11$ and common ratio $r = -\frac{1}{4}$, using $S = \frac{a_1}{1-r}$.

a. $\frac{55}{4}$

b. $\frac{44}{5}$ ✓

c. -44

d. $\frac{44}{3}$

$S = \frac{a_1}{1-r} = \frac{11}{5/4} = \frac{44}{5}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Statistics, Counting & Probability

In an arithmetic sequence, the 4th term is 19 and the 11th term is 54. What is the first term a_1?

a. 4 ✓

b. 5

c. 6

d. 7

Common difference $d = \frac{a_{11} - a_4}{11 - 4} = \frac{54 - 19}{7} = \frac{35}{7} = 5$. Then $a_1 = a_4 - 3d = 19 - 3(5) = 19 - 15 = 4$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Advanced Number Theory

What is the 4th term in the expansion of $(x + 2)^5$ arranged in descending powers of x?

a. $40x^2$

b. $80x^2$ ✓

c. $20x^3$

d. $10x^3$

4th term has $k = 3$: $\binom{5}{3}x^{5-3}(2)^3 = 10x^2(8) = 80x^2$.

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